CHRISTOPHER MARONGA

Statistics, Data Management & R Programming

SKILLS

R	10+ yrs
REDCap	6+ yrs
SQL/MySQL	5+ yrs
Git/GitHub	4+ yrs
Phase III & IV CTs	3+ yrs
LaTeX	2+ yrs
Real World Data	1+ yrs
Python	< 1 yr
CDISC(SDTM , ADaM)	< 1 yr

EDUCATION

Sep 2019 - Dec 2021

MSc. in Statistical Science
Strathmore University

May 2009 - Jun 2013

B.Sc. in Mathematics
Jomo Kenyatta University of Agriculture and Technology

PROFILE

Clinical Data Manager and Biostatistician with extensive experience in Real-World healthcare Data, observational epidemiology, and statistical modelling. Currently managing large-scale NHS-linked cohort data at the University of Oxford, delivering secondary analyses of electronic health for chronic disease outcomes.

Experienced in SQL-based data extraction, R programming, survival analysis (Cox regression, Kaplan-Meier), and multivariable modelling. Proven ability to independently design analytical solutions for complex healthcare datasets and collaborate cross-functionally to generate robust real-world evidence supporting clinical decision-making.

WORK EXPERIENCE

University of Oxford - Clinical Data Manager Sep2023 - present
Oxford Population Health, Cancer Epidemiology Unit

Designing and management of large-scale longitudinal research databases within the Cancer Epidemiology Unit, Oxford Population Health. Leading database infrastructure for a nationally linked cohort study (65,000–145,000 participants), supporting epidemiological research through advanced data management, linkage with NHS datasets, and collaboration with statisticians and researchers.

- Design and implement SQL-based data extraction pipelines for observational epidemiological studies.
- Develop reproducible analytical workflows in R for longitudinal and real-world healthcare data.
- Perform secondary analysis of large-scale NHS-linked electronic health records and cohort datasets.
- Collaborate with epidemiologists and statisticians to design and validate study methodologies.
- Ensure data governance compliance (GDPR) and secure data sharing processes.

Technologies:

- Electronic Health Records linkage
- PostgreSQL

Achievements:

- Developed automated survey invitation and response workflows using R, reducing manual workload and improving response turnaround time.

CONTACT

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Website

University of Oxford - Data Manager May2023-Aug2024

Oxford-GSK Institute for Molecular and Computational Medicine (IMCM)

Provide data management and scientific computing capabilities to support IMCM projects. Apply informatics and data management skills to help identify underlying mechanisms and translate knowledge into development of new, precision therapeutics in neurodegenerative diseases.

- Develop and use novel methodologies for storing information and datasets along with their metadata.
- Ensure datasets are appropriately annotated and made available in a FAIR (Findable, Accessible, Interoperable, Reusable) manner.
- Develop and implement operational procedures that underpin Information Governance policies, working closely with the NDM & GSK IG teams.
- Produce end-user documentation and training on data management processes.
- Provide data management input into outline and full grant proposals, as co-applicant.
- Work with GSK teams to ensure that data sets are made available in a timely and self-documenting fashion.

Technologies:

- Familiarity with OHDSI - OMOP CDM
- R/Linux/Docker

Achievements:

- Authored an R package to aid data accessibility in Terra.Bio cloud platform

University of Oxford - Data Analyst (R Programmer) Jun 2021-May 2023

Nuffield Department of Orthopaedics, Rheumatology and Musculoskeletal Sciences

Working with the Health Economics and Outcome Research Group to develop and apply innovative algorithms/systems to extract knowledge and insights from health care data

- Extract, clean, validate, investigate, prepare, and report healthcare-related data using R
- Responsible for developing and maintaining common library of R functions for team collaboration
- Responsible for researching & applying statistical methods, build models when needed.
- Contribute in developing manuscripts for publication.
- Contribute in developing joint R packages specific to health economics
- Build interactive web apps and dashboards using R-Shiny for dynamic reporting and data curation
- Responsible for reviewing code written by other researchers, help troubleshoot and suggest efficient ways of coding

Technologies:

- Rshiny
- Rmarkdown

Achievements:

- Built a fully automated reports aggregation for a micro-simulation model using Rshiny & Rmarkdown

Clinical Data Manager Oct 2017-May 2021

KEMRI-Wellcome Trust Research Programme,

The CHAIN Network is a large multi-continent, multi-country cohort study with a total of 9 sites across Africa and South Asia

- Manage large-scale clinical and observational research databases across multi-site studies.
- Designed relational database systems and electronic data capture tools (REDCap).
- Led data quality assurance, validation, and harmonisation across multiple study platforms.
- Play a central role in ethical data sharing within the network researchers
- Data systems infrastructure maintenance and troubleshooting
- Supported statistical analyses including survival modelling and longitudinal outcome assessments.

Technologies:

- Rshiny
- Tidyverse

- Rstudio Server
- Shiny Server
- Git for collaboration and documentation versioning

Achievements:

- Conceived and developed a robust Shiny applicaton for automated data cleaning, query resolution, report authoring, routine statistical analysis and visualizations.
- Boosted data managers productivity through hosting automated customized sites weekly reports on Shiny dashboard

Technical Advisor – Data Management Jan 2017-Sep 2017

Partners PrEP Scale-Up Project (KEMRI CCR Project)

An implementation project to scale-up delivery of antiretroviral-based HIV-1 prevention methods to serodiscordant couples in 24 public HIV-1 care centers in central and western Kenya

- Provide technical oversight for healthcare data systems within HIV implementation research.
- Prepare and submit to coordinating centre a cumulative report of selected clinics' indicators on a monthly basis for evaluation of project progress
- Design data collection frameworks and ensure regulatory-compliant data handling.
- Support analysis of routine healthcare data to evaluate programmatic outcomes.
- Monitor study accrual rates and generate QC reports for the 13 participating sites and follow up to ensure a timely resolution of the data queries

Technologies:

- SurveyCTO
- R software

Achievements:

- Harmonized and centralized key indicator reporting for clinics using surveyCTO and automatically aggregate the reports for consumption.

Assistant Data Manager Jul 2014-Jan 2017

KEMRI-Centre for Clinical Research (KEMRI CCR)

Enviroment of multi-Clinical Trials and Research studies and invovled in over 5 clinical trails data management activities

- Designing of REDCap databases for participants data storage
- Preparation of daily, weekly and monthly study data summaries to track and inform the trials' progress

- Perform data entry of participants' records into REDCap data base
- Configure the SCIP application on tablets for SCIP Msurvey trials
- Design of data capturing systems/tools, development of SOPs for data recording and entry, and case report forms for data collection.
- Data QA/QC review on REDCap, validation and discrepancy checking
- Perform regular cloud data backup for the study, develop comprehensive system logs in excel to track data query work flow
- Co-ordinate ordering/procuring and distribution of both hardware & software and keep up to date the site's inventory for IT equipment
- Maintain and update on a daily basis participant's master data base, which contains client's details, progress through the trial, adverse events and other relevant research related information
- Closely working with study monitor to oversee quality assurance and quality control procedures for the clinical trial
- Training and supervising of research assistants, interns and data entry clerks at the Thika site
- Maintain systematic data storage of participants' data (both electronic and binder files) as per Good Clinical Practice(GCP).

Technologies:

- Ms Access
- REDCap
- R for reporting and data management

Achievements:

- Developed a robust system using Ms Access and Ms Excel for automated patient follow up tracking and report generation on missed appointments and lost to follow up
- Design over 6 usable databases both in REDCap and Ms Access

Research Assistant(Intern) Oct 2013-Jun 2014
KEMRI-Centre for Clinical Research (KEMRI CCR)

A project to evaluate feasibility and potential impact of text messages on HIV awareness in young women in rural Kenya. Tasks included

- Study coordinator, strategic planning of participant recruitment into the study and short messages schedules with msurvey Kenya, performed data cleaning and preliminary data analysis and report generation using SPSS
- Designed storage data base and managed data entry, data cleaning and regular survey data back-up
- Assist t2t PI in drafting project proposal, abstracts, publications and budgets
- Responsible for confidential storage and documentation of study data and link logs according to GCP
- Preparations of bi-weekly and weekly data summaries as required for study retention and follow up to assess and inform status/progress of t2t study project

- Perform any duty as assigned by t2t Principal Investigator, Site coordinator and Lead Clinician.

Technologies:

- Ms Access
- SPSS for reporting

Achievements:

- Oversaw data management for the study from start to successful completion with a high quality data output

Publications

Expected Benefits and Budget Impact From a Microsimulation Model Support the Prioritization and Implementation of Fracture Liaison Services

Primary care consultations and pain medicine prescriptions: a comparison between patients with and without chronic pain after total knee replacement

Childhood mortality during and after acute illness in Africa and south Asia: a prospective cohort study

A systematic review of the quality of conduct and reporting of survival analyses of tuberculosis outcomes in Africa

The Effect of Human Immunodeficiency Virus Prevention and Reproductive Health Text Messages on Human Immunodeficiency Virus Testing Among Young Women in Rural Kenya

The Incremental Cost of Delivering PrEP as a Bridge to ART for HIV Serodiscordant Couples in Public HIV Care Clinics in Kenya

Talks & Workshops

NHS-R Workshop: Working with databases in R

Innovative open source approaches to automating clinical data management for a large multi-centre cohort study - REDCap & RShiny

satRday Nairobi - Introduction to Data Management in R

NairobiR & R-Ladies Webinar - Working with databases in R

Technical Skills

- Advanced R Programming
- Statistical analysis (Time to Event & Longitudinal data analysis)
- Relational Databases (PostgreSQL/MySQL) & advanced SQL querying

References

Dr. Moses Ngari
KEMRI-Wellcome Trust Research Programme
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